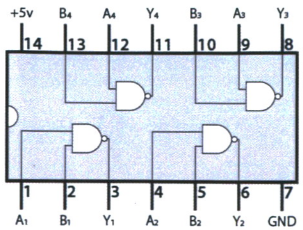
Integrated circuit quick reference

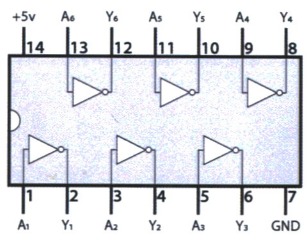
74LS00: NAND gate

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0



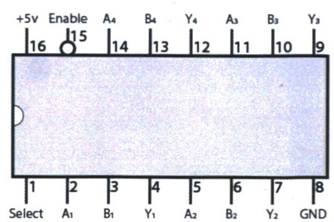
74LS04: Inverter

A	Y
0	1
1	0



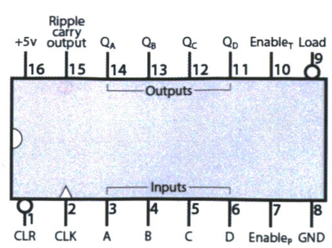
74LS157: 2-to-1 line selector

A 4-bit input is selected from one of two sources (A or B) and is routed to the four outputs (Y).



74LS161: 4-bit binary counter

The 4-bit output is incremented on each CLK pulse. Alternately, any value can be loaded via the 4-bit input.



If you get stuck

There's no denying it: Building a computer from scratch on breadboards is a big project. The best part is that by the time you're done, you'll have an incredible understanding of how the computer works. But at least some of that understanding will come because you had to track down some (possibly frustrating) problems. The good news is you're now part of a community of people working on this project and helping each other out. There are several good places to look for help:

- Comment threads on the YouTube videos.
- Comment threads at eater.net/8bit.
- Questions and discussions at reddit.com/r/beneater.

Caution:

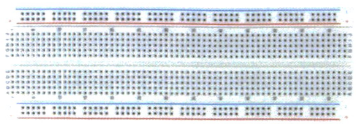
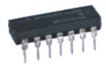
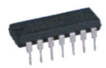
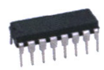
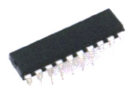
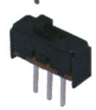
You should never connect LEDs directly to a 5 volt source without a current limiting resistor. Doing so will damage the LED.

The outputs of most 74LS*xx* logic chips have internal resistors that let you to connect LEDs directly to the outputs without a resistor. (The 555 timer, 74189 RAM, 28C16 EEPROM, and 74HC595 *do not.*)

For the most reliable results, you may want to connect a 220Ω resistor in series with every LED anyway. I omitted many of these as a shortcut in the videos, but I've included enough 220Ω resistors in these kits if you choose (or need) to use them.

Components included in this kit

Note that some components might look slightly different than shown here or in the videos.

4	Breadboards		
1	74LS00 quad NAND gate		
2	74LS04 hex inverter		
4	74LS157 quad 2-1 line mux		
1	74LS161 4-bit binary counter		
1	74LS173 4-bit D register		
2	74189 64-bit RAM		
2	74LS245 8-bit bus transceiver		
1	Momentary pushbutton		
1	Slide switch	This replaces the push-button toggle switch used in the videos.	

Program counter schematic